

Conditional units and force paths

Exact algebra checked. Physical specimen calibration remains missing.

1 Physical normalizers

$C = T_{ref} \times F_{scale} \quad [N]$
 $\ell = L_{ref} / \Gamma \quad [m]$
 Γ : dimensionless
 $\gamma = 2 \text{ Ns dps}$: length [m]

Missing paired force / area / tare
and reference length ledger

2 Compatible elastic circuit

Smooth PE prefactor: $C \text{ kpe} / K \quad [N]$
PE width: $\ell / K \quad [m]$
PE slope: $C \text{ kpe} / \ell \quad [N/m]$
SE prefactor: $C \text{ kse0} \quad [N]$
SE exponent: $\text{kse} / \ell \quad [1/m]$

Smooth PE branch only
Published parity unresolved

3 Area and force topology

Parallel: $T = H \text{ kCB dps } Q$
Serial: adds length and energy
Serial links transmit the same force
Cauchy = $(\lambda / J) \times \text{nominal}$

PCSA vs projected FCSA:
cosine applied once

What the audit establishes

12 exact identities • 4 intended dimensional rejections

2 unit-consistent mechanical errors exposed: serial-force multiplication and double pennation projection

No empirical calibration, kinetic ODE, tissue solve or convergence comparison.